

MAST32001 Computational Statistics

Exercise Sheet 5 – Markov chains and MCMC

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Submission

Solutions must be submitted in **Moodle no later than 07.10.2026 at 12:00**.

Please submit your solutions in the **submission area corresponding to your exercise group**. Make sure that you submit to the correct exercise group.

Exercise points

Exercise points are awarded only in connection with the corresponding exercise session.

To receive the point(s) for a task, you must:

1. submit your solution by the deadline;
2. attend the exercise session in which the task is discussed; and
3. be willing to **present, explain, and discuss your solution** during the exercise session if asked.

Submitting a solution without attending the corresponding exercise session does **not** earn exercise points.

Your solution does not need to be perfect in order to be discussed. The purpose of the exercise sessions is also to compare approaches, identify difficulties, and learn from different solutions.

Use of generative AI

The use of **generative AI tools to solve the exercise tasks is not permitted**.

The purpose of the exercises is to develop your own ability to translate statistical ideas into algorithms, implement them in R, investigate numerical behaviour, and explain your reasoning.

The submitted solutions, including the reasoning and R code, must therefore reflect **your own work**.

Exercise 1: A Markov chain with circulation

Consider

$$P = \begin{pmatrix} 0.20 & 0.65 & 0.10 & 0.05 \\ 0.05 & 0.20 & 0.65 & 0.10 \\ 0.10 & 0.05 & 0.20 & 0.65 \\ 0.65 & 0.10 & 0.05 & 0.20 \end{pmatrix}.$$

1. Find a stationary distribution π , and explain how you obtained it.
2. Is the chain reversible with respect to π ?
3. Is it irreducible and aperiodic? Justify both answers.
4. Starting from state 1, simulate 10,000 steps in R and compare empirical state frequencies with π .
5. For $h(x) = I(x \in \{1, 2\})$, plot the running average

$$\frac{1}{m} \sum_{t=1}^m h(X_t), \quad m = 1, \dots, 10000,$$

and add its theoretical limiting value. Interpret the plot.

Exercise 2: Metropolis–Hastings on a transformed scale

Suppose

$$Y \mid p \sim \text{Binomial}(40, p), \quad Y = 31.$$

Define

$$\eta = \log\left(\frac{p}{1-p}\right)$$

and assume directly that

$$\eta \sim N(0, 2^2).$$

1. Derive the log posterior of η , up to an additive constant.
2. Implement random-walk Metropolis with

$$\eta^* = \eta^{(t)} + \varepsilon, \quad \varepsilon \sim N(0, s^2).$$

3. Run 20,000 iterations with $s = 1$, starting at zero, and discard the first 2,000.
4. Transform the retained draws to p and estimate $E(p \mid y)$ and $P(p > 0.75 \mid y)$.
5. Explain why the logit scale is convenient compared with a Gaussian random walk directly on p .

Exercise 3: Diagnose and repair a sampler

The target is $N(0, 2^2)$. Use the following sampler.

```
log_target <- function(x) dnorm(x, 0, 2, log=TRUE)

rw_normal <- function(initial, n_iter, proposal_sd) {
  x <- numeric(n_iter); x[1] <- initial; acc <- 0
  for (i in 2:n_iter) {
```

```

prop <- rnorm(1, x[i-1], proposal_sd)
if (log(runif(1)) <
    log_target(prop)-log_target(x[i-1])) {
  x[i] <- prop; acc <- acc+1
} else x[i] <- x[i-1]
}
list(draws=x, acceptance=acc/(n_iter-1))
}

```

1. Run four chains of length 2,000 from $-12, -4, 4, 12$ using `proposal_sd = 0.03`.
2. Produce suitable diagnostic plots and numerical summaries. Decide yourself what is useful.
3. Explain what is wrong and why the acceptance rate alone is misleading.
4. Modify the simulation to obtain substantially better output. Justify the change and demonstrate the improvement.
5. Compare estimates of $E(X)$ and $\text{Var}(X)$ with their known values.

Exercise 4: Multimodality

Consider

$$\pi(x) = \frac{1}{2}\phi(x; -5, 1) + \frac{1}{2}\phi(x; 5, 1).$$

1. Implement the log target. Avoid unnecessary numerical instability when evaluating the mixture.
2. Run four random-walk Metropolis chains of length 15,000 with proposal SD 0.8, starting near both modes.
3. Plot the chains and calculate the fraction of retained draws greater than zero for each chain.
4. Explain why a single apparently stable trace plot can be seriously misleading here.
5. Modify the proposal mechanism to improve movement between modes. Demonstrate that your modification helps. More than one solution is possible.

Exercise 5: MCMC as numerical integration

Consider the unnormalized density

$$\tilde{\pi}(x) = \exp\left(-\frac{x^4}{4} + \frac{x^2}{2}\right)$$

and

$$\mu = E_{\pi}\left\{\frac{1}{1+X^2}\right\}.$$

1. Explain why Metropolis–Hastings does not require the normalizing constant.
2. Use `integrate()` to obtain a high-accuracy numerical benchmark for μ .
3. Implement random-walk Metropolis. Choose a proposal scale using diagnostics rather than simply reporting the first value tried.
4. Estimate μ from retained MCMC draws.
5. Estimate the Monte Carlo standard error in a way that accounts for serial dependence. Compare it with the actual difference from the numerical benchmark. Explain why `sd(h(x))/sqrt(M)` is generally inappropriate for MCMC output.